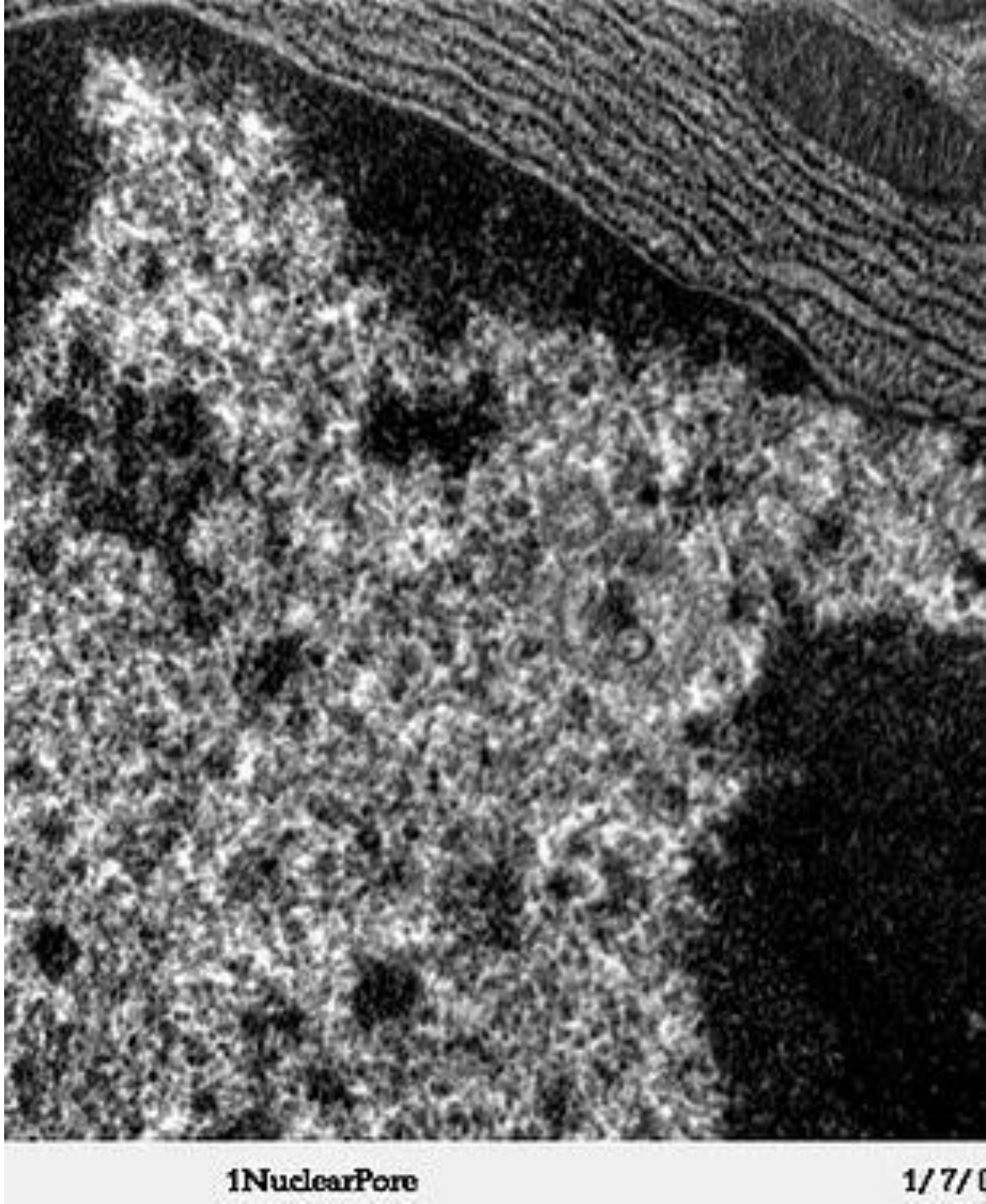
DP Biology – B2.2

Organelles and compartments

- How does the structure of each organelle suit its function?
- Why is it useful for cells to have separate compartments?

SL statements B2.2.1–3. Textbook pp. 244–248.
Image: nucleus and rough ER in a pancreas cell (TEM). Public domain.

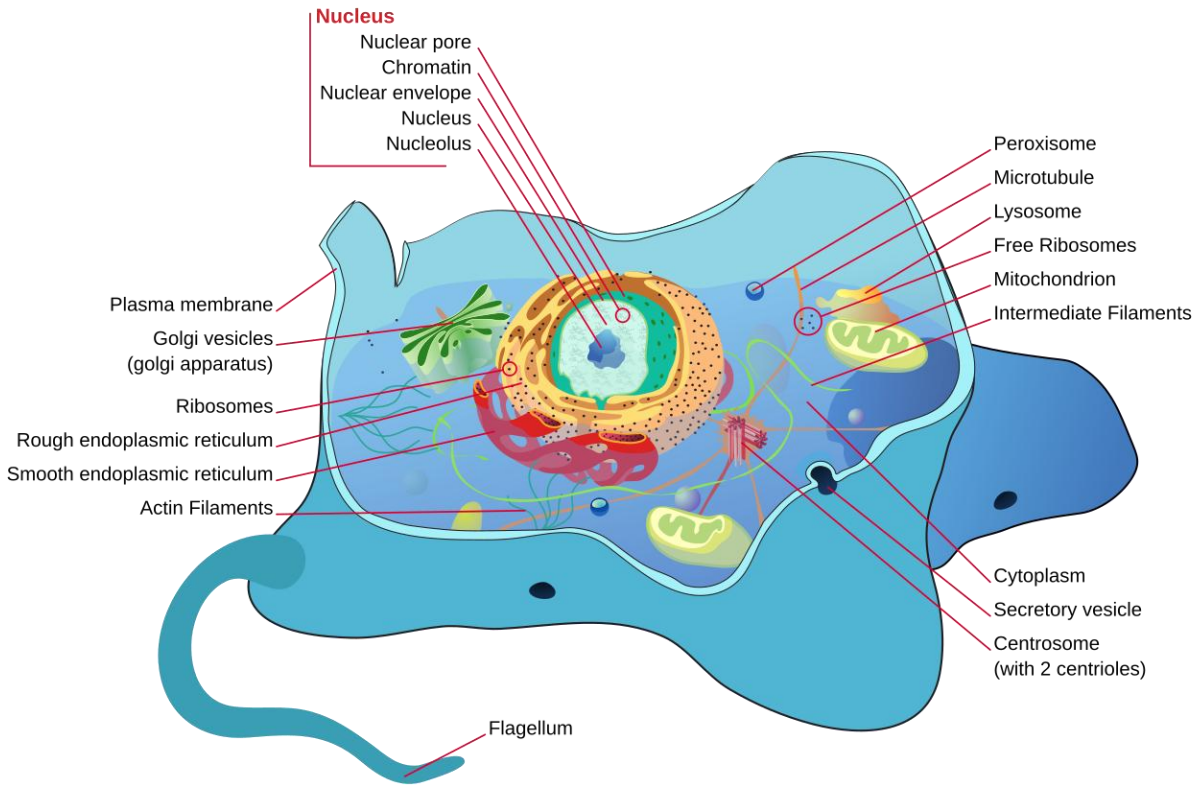


Objectives for each class

Class	Statements	By the end of the class, you can…
1	B2.2.1	Define an organelle and sort cell structures by their membranes, with structure linked to function
2	B2.2.2–3	Explain the advantages of separating the nucleus from the cytoplasm, and of compartments in the cytoplasm

A cell is like a factory

B2.2.1–3



An animal cell. LadyofHats, public domain.

A factory has separate rooms: an office for the plans, workshops, a packing room and a waste area.

A eukaryotic cell works the same way. Its **organelles** are the rooms, each with its own job and conditions.

This chapter

What counts as an organelle, and why keeping things apart helps.

What is an organelle?

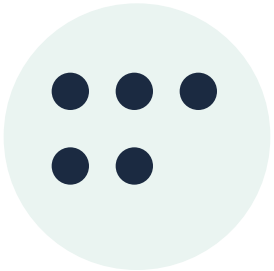
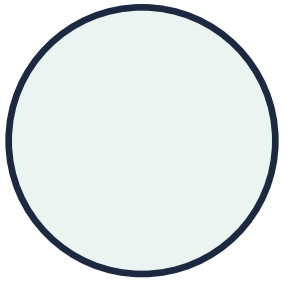
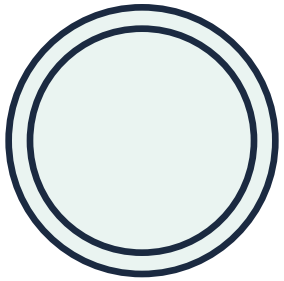
B2.2.1

An organelle is a separate part of a cell, adapted to carry out a particular job.

Organelles	Not usually called organelles
nucleus, mitochondria, chloroplasts	cell wall: outside the membrane
ER, Golgi, vesicles, lysosomes, vacuoles	cytoskeleton: spread through the whole cell
ribosomes	cytoplasm: the fluid around everything

Sorting by membranes

B2.2.1

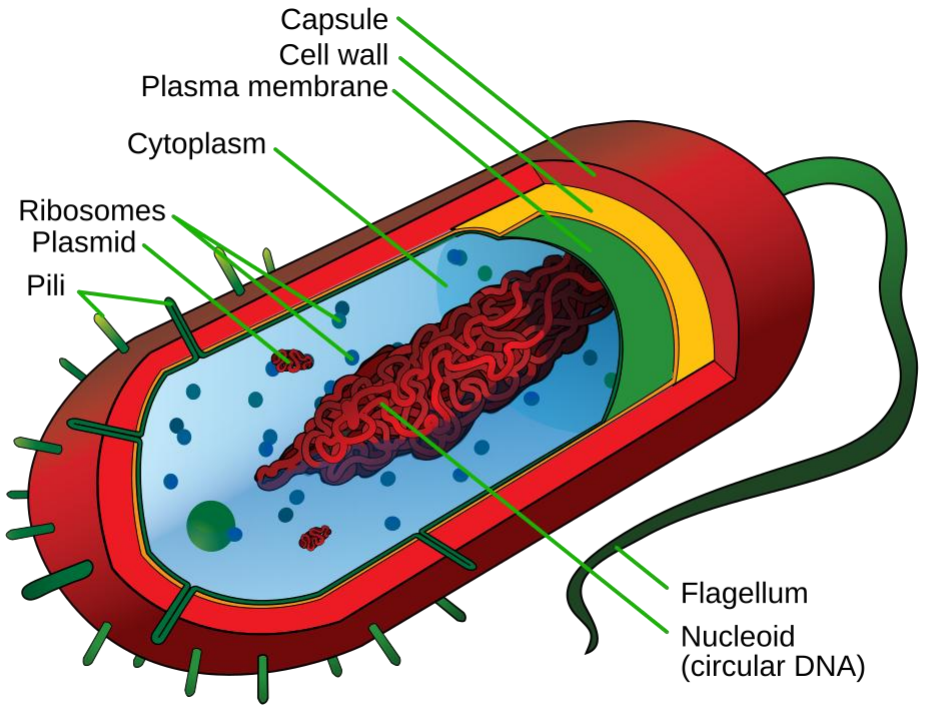
No membrane	One membrane	Two membranes
		
ribosomes, centrioles	ER, Golgi, lysosomes, vesicles, vacuoles	nucleus, mitochondria, chloroplasts

Link to HL

Mitochondria and chloroplasts have two membranes because they came from bacteria engulfed by another cell (endosymbiosis, A2.2 HL).

Prokaryotes: few organelles

B2.2.1



A prokaryote. LadyofHats, public domain.

Bacteria have **no membrane-bound organelles**: no nucleus, no mitochondria.

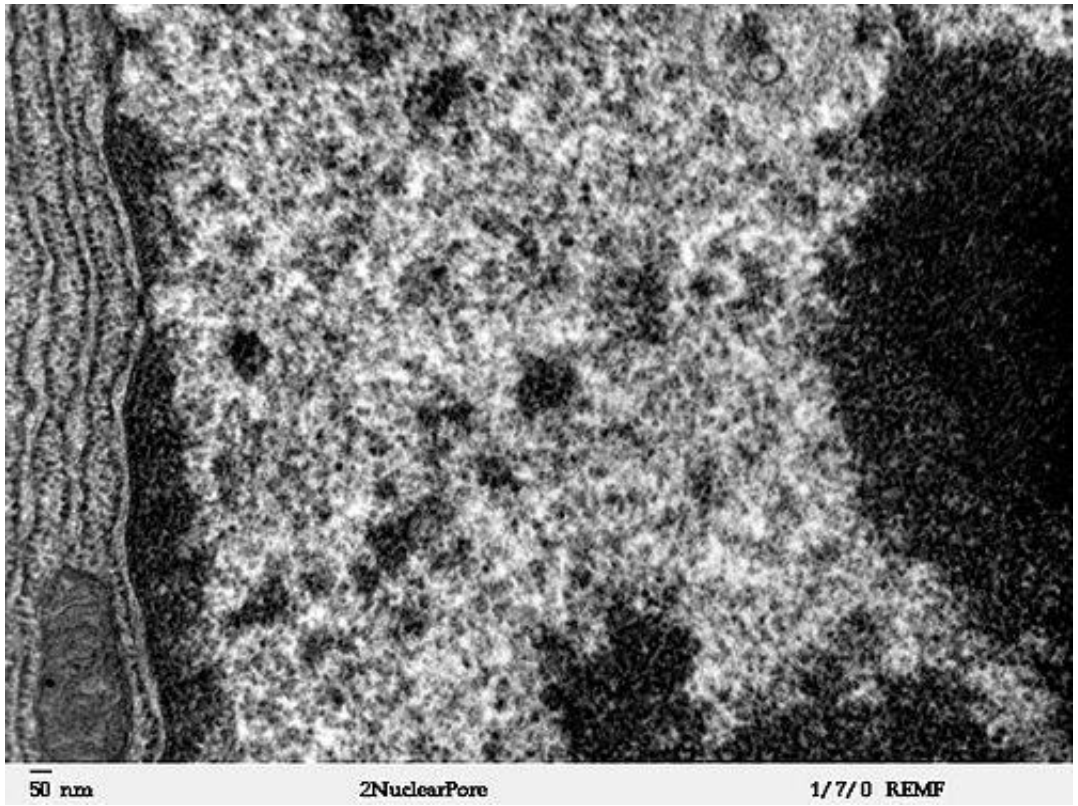
Their only organelles are **ribosomes**. Everything else happens in one undivided cytoplasm.

Compare

Prokaryotes are small and simple; eukaryotes are larger and compartmentalized.

The nucleus

B2.2.1–2



Nuclear envelope with pores (TEM). Public domain.

A **double membrane** (the nuclear envelope) surrounds the DNA.

Nuclear pores let mRNA out and proteins in.

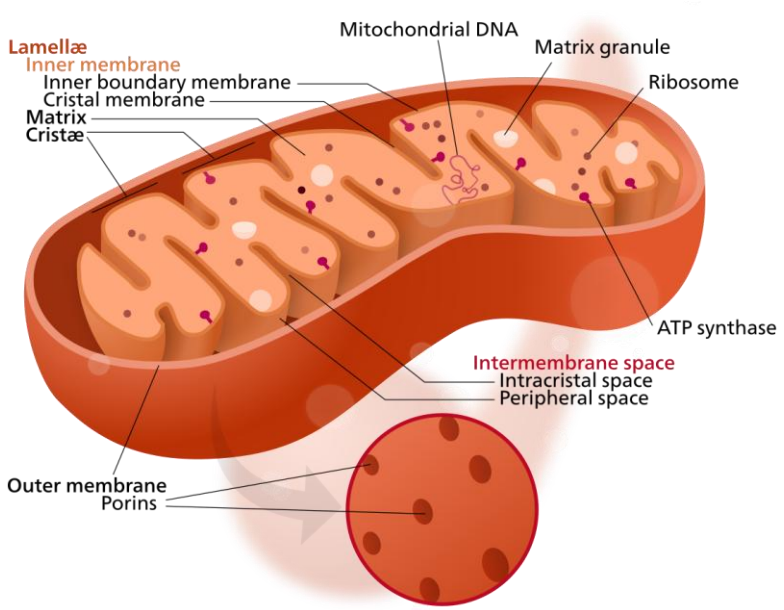
The nucleus stores the genetic instructions and is where **transcription** happens.

Look

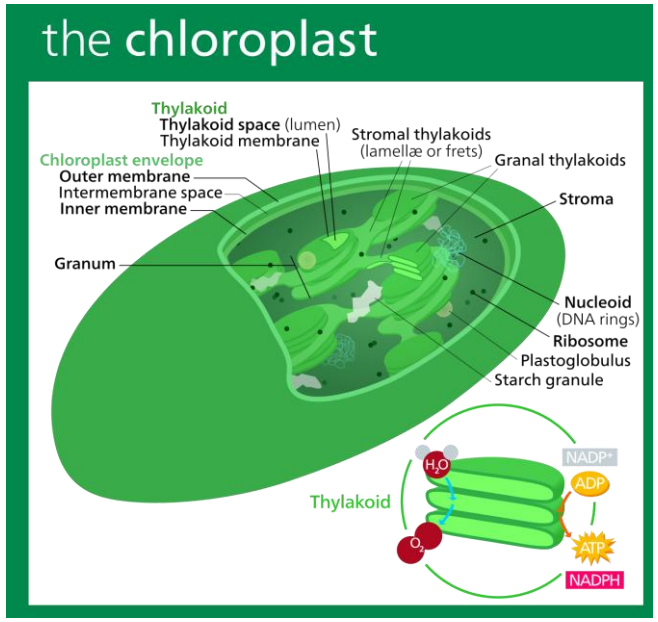
In the micrograph, find the two membranes and the gaps where pores cross them.

Two organelles that release and capture energy

B2.2.1



Mitochondrion. CC BY-SA 3.0.



Chloroplast. CC BY 3.0.

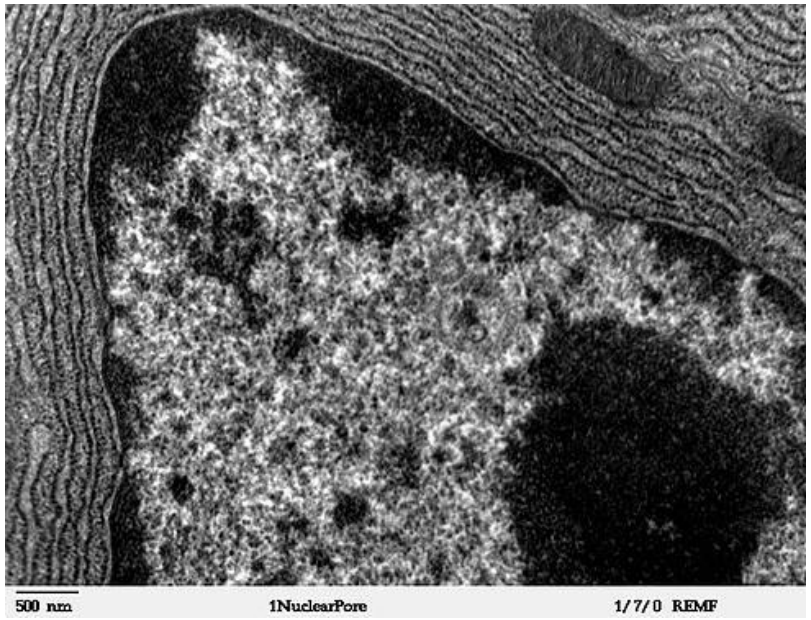
Mitochondrion

Aerobic respiration makes ATP. The inner membrane is folded into cristae, giving a big area.

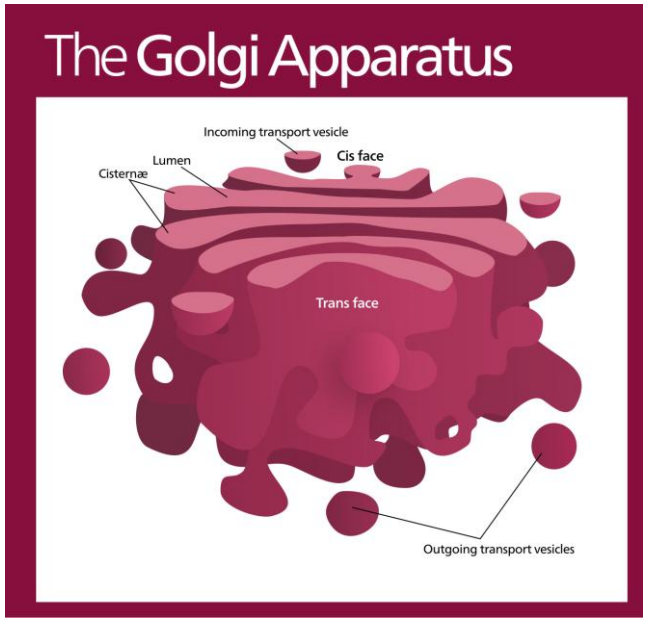
Chloroplast

Photosynthesis. Stacks of thylakoid membranes hold the chlorophyll.

ER and Golgi apparatus



Rough ER (TEM). Public domain.



Golgi apparatus. CC BY 3.0.

Rough ER: membranes covered in ribosomes; makes proteins for export.

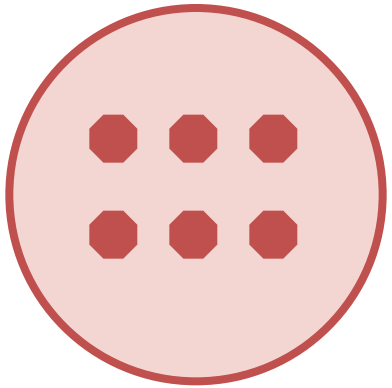
Smooth ER: makes lipids.

Golgi: stacks of sacs that modify and pack proteins into vesicles.

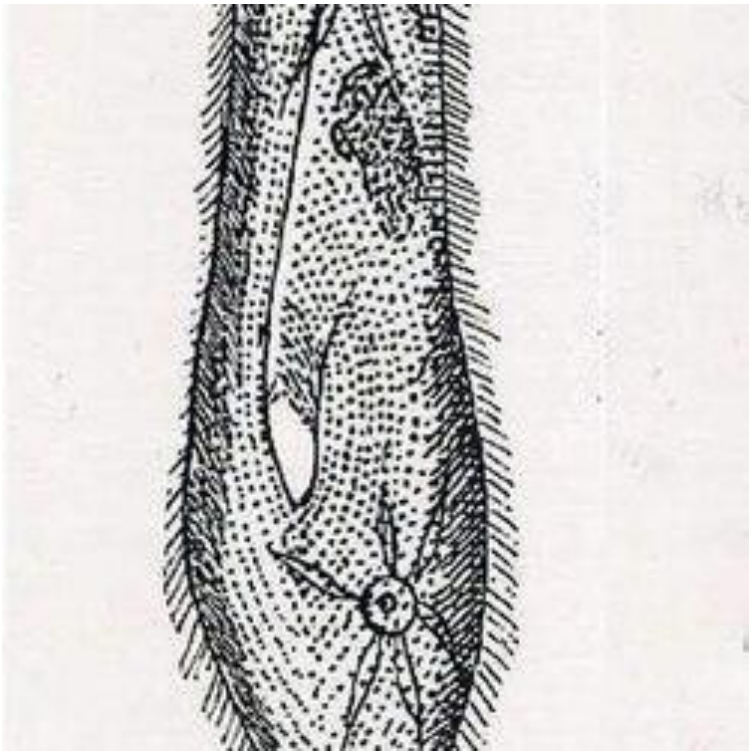
Look

In the micrograph, the dark dots along the membranes are ribosomes: this is rough ER.

Lysosomes and vacuoles



lysosome



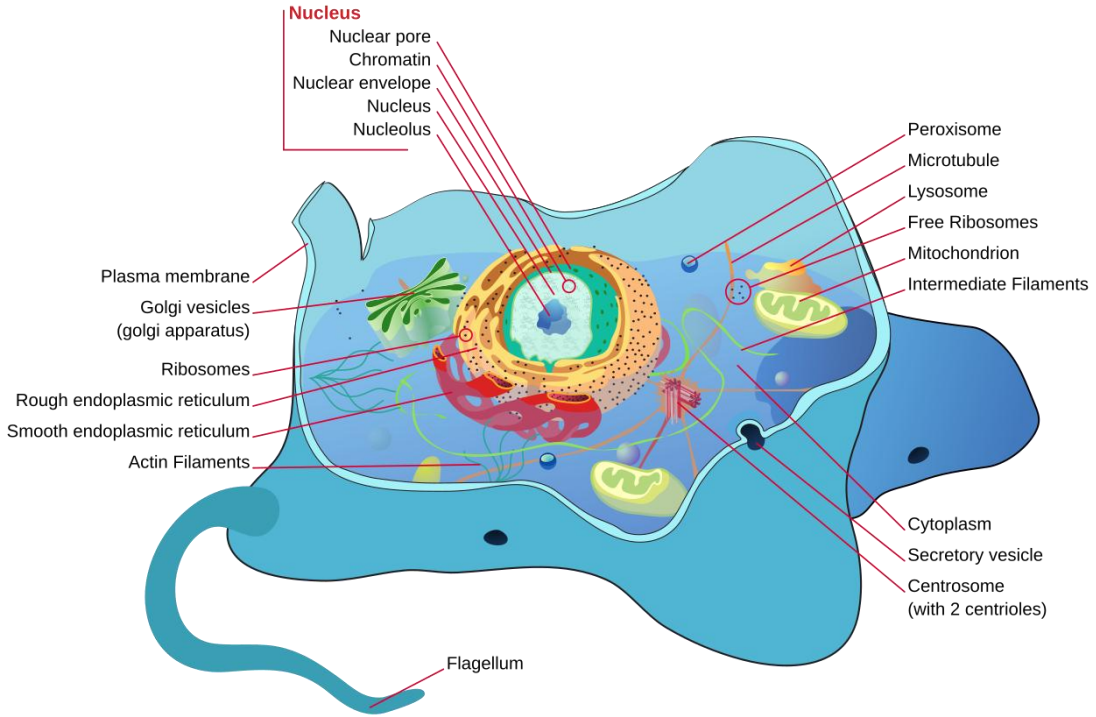
Paramecium with food vacuoles. Public domain.

Lysosomes hold digestive enzymes that break down worn-out organelles and food taken in.

Vacuoles store water and substances; in a protist such as Paramecium, **food vacuoles** digest food.

Practice: name and sort

B2.2.1



Animal cell. LadyofHats, public domain.

1. Pick five structures in the diagram and state the job of each.
2. Sort them into no membrane, one membrane and two membranes.
3. Name one structure that is not an organelle, and explain why.

Working with data: separating organelles

B2.2.1

Cells were broken up and spun in a centrifuge. Enzyme activity in each fraction (% of total). These data are invented for practice.

Fraction	Enzyme X / %	Enzyme Y / %
Nuclei	4	2
Mitochondria	88	6
Membranes (ER, Golgi)	5	81
Liquid left	3	11

1. Identify where enzyme X works.
2. Suggest a job for enzyme Y.
3. Explain why every fraction shows a little activity.

Check-in: class 1

B2.2.1

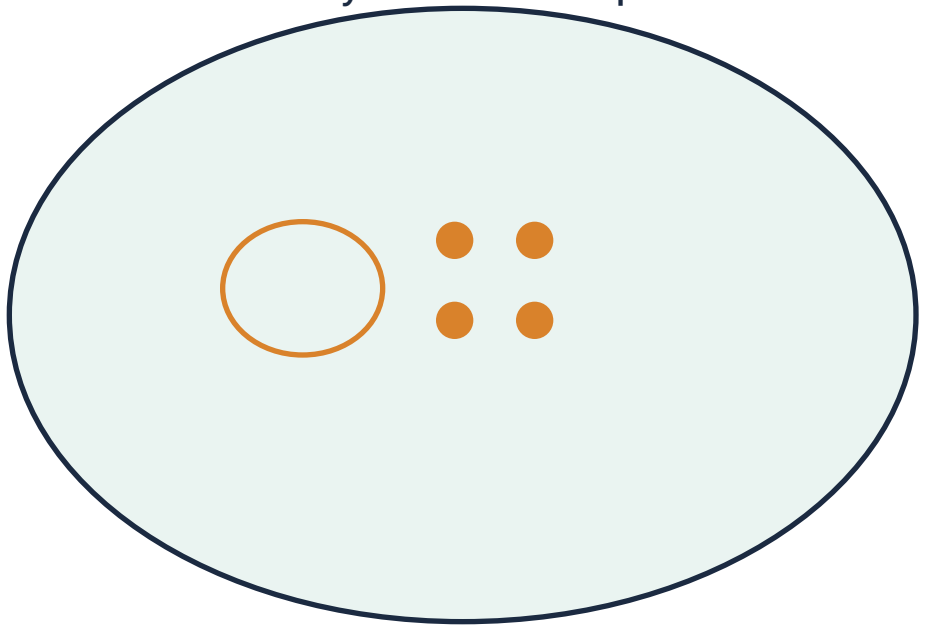
1. Define an organelle.
2. Give two structures with no membrane, two with one and two with two membranes.
3. Explain why the cell wall is not called an organelle.
4. State the job of the nucleus, rough ER, Golgi and lysosomes.
5. State the only organelles found in prokaryotes.

How to use this

Answer without your notes, then check against slides 4–10.

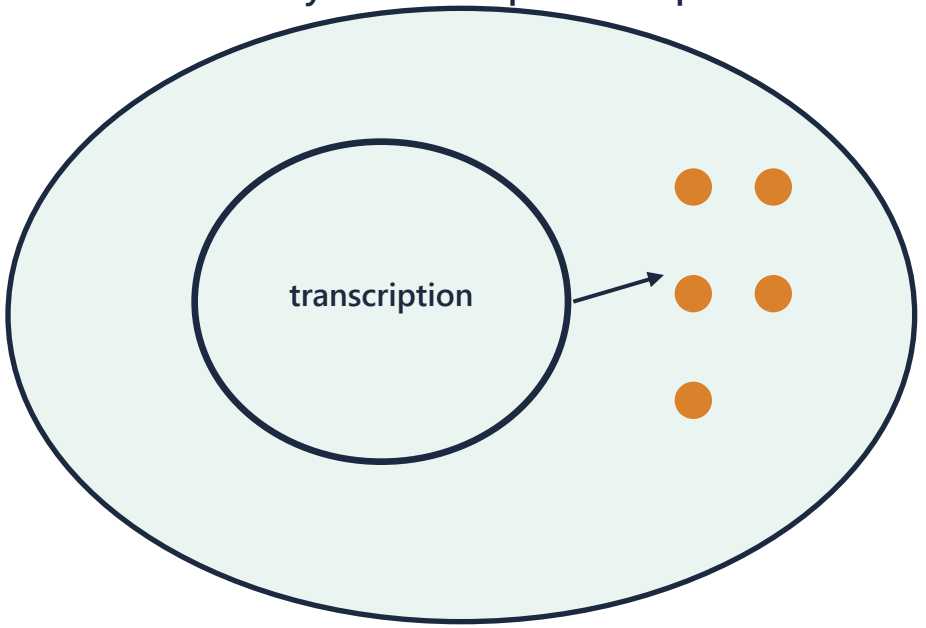
Why keep the nucleus separate?

Prokaryote: all in one space



translation starts while mRNA is still being made

Eukaryote: two separate steps



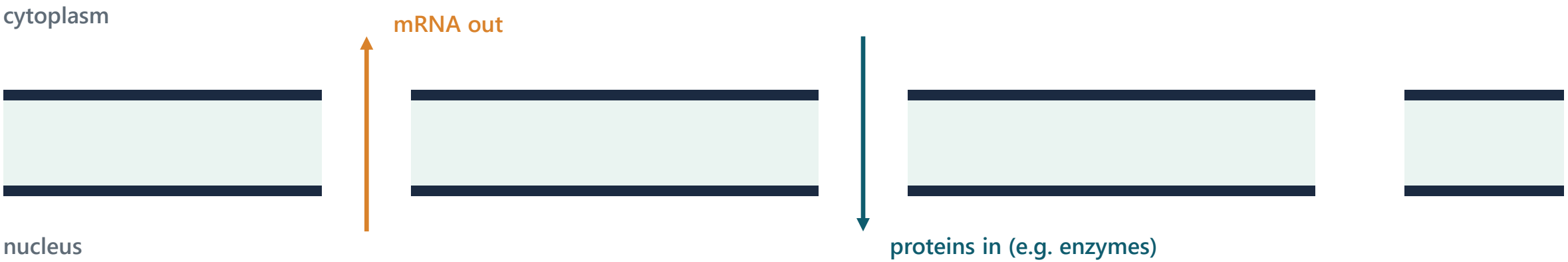
translation

In a eukaryote, mRNA is made in the nucleus but translated outside it. This gap in space and time lets the cell **edit the mRNA** (remove introns) before ribosomes can reach it.

It also protects the DNA from damage by reactions in the cytoplasm.

Nuclear pores: controlled traffic

B2.2.2



Pores are not just holes: they decide what passes. mRNA goes out to the ribosomes; proteins needed for copying DNA and transcription come in.

Why divide up the cytoplasm?

B2.2.3

Concentrate — enzymes and substrates kept together, so reactions are fast

Separate — harmful enzymes (e.g. in lysosomes) kept away from the rest of the cell

Different conditions — each compartment can have its own pH, like acidic lysosomes

Specialize — many jobs can run at once without getting in each other's way

Example

Lysosomes are at pH about 5; the cytoplasm is about 7.2.

A cell's job decides its organelles

B2.2.1, 3

Pancreas cell	Muscle fibre	White blood cell	Leaf palisade cell
lots of rough ER and Golgi	many mitochondria	many lysosomes	many chloroplasts
makes and secretes digestive enzymes	needs constant ATP to contract	digests the bacteria it engulfs	photosynthesis

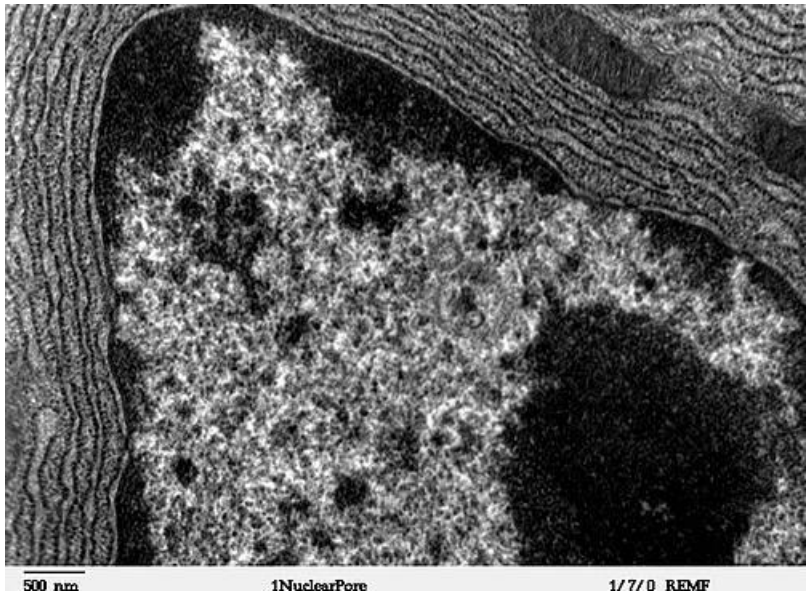
Key idea

All these cells have the same kinds of organelle, but in very different numbers. Count the organelles in a micrograph and you can often guess what the cell does.

Teamwork: making a protein for export

B2.2.3

Nucleus: gene transcribed → Rough ER: protein made and folded → Vesicle carries it → Golgi: modified and packed → Vesicle fuses with membrane: protein released

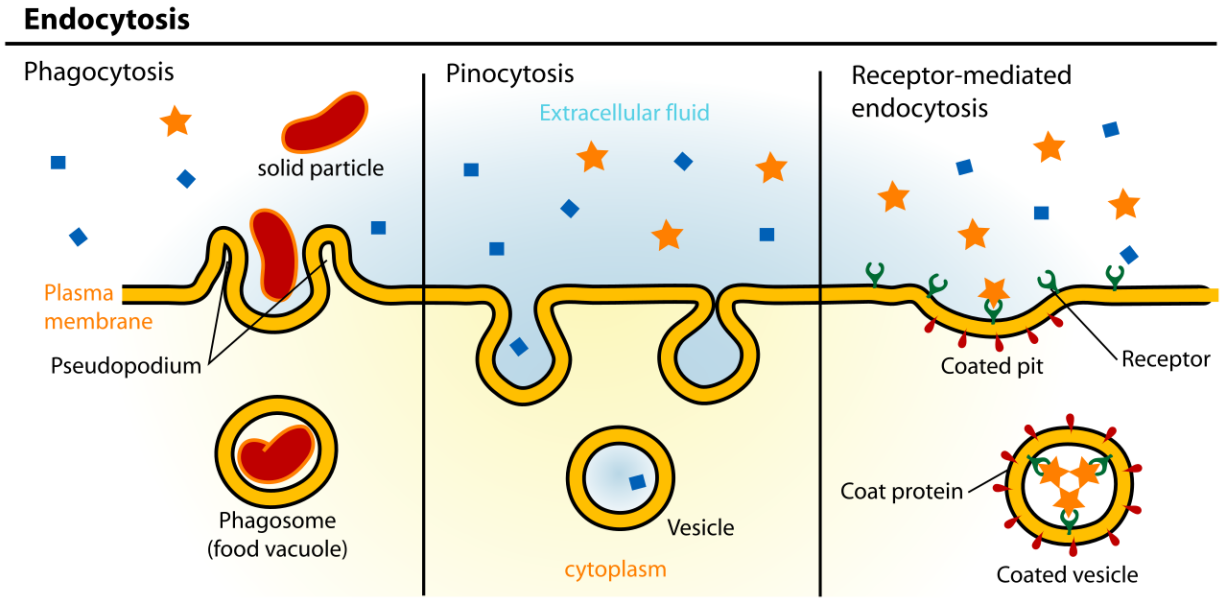


Rough ER (TEM). Public domain.

Each compartment does one step, and **vesicles** carry the protein between them. Digestive enzymes and hormones such as insulin are made this way.

Compartments in action: digesting food

B2.2.3



Endocytosis. Public domain.

Because the digestion happens inside a compartment, the enzymes never touch the rest of the cell. White blood cells destroy bacteria in exactly this way.

Cell engulfs food in a vesicle → Lysosome fuses with it → **Enzymes digest the food** → Useful molecules absorbed

Practice: what would go wrong?

B2.2.2–3

A lysosome bursts

Predict the effect on the cell.

Nuclear pores block

Explain what would happen to protein synthesis.

No ER or Golgi

Suggest why a pancreas cell could not release enzymes.

How to use this

Answer each in two sentences using the words compartment and enzyme.

Working with data: lysosome enzymes and pH

B2.2.3

Activity of a lysosomal enzyme at different pH. These data are invented for practice; the pattern matches real enzymes.

pH	Activity / %
4	80
5	100
6	45
7	8
8	2

1. State the optimum pH.
2. Explain why the enzyme does little harm if it leaks into the cytoplasm.
3. Explain how this shows an advantage of compartments.

Common mistakes to avoid

Not this	But this
Ribosomes are surrounded by a membrane.	Ribosomes have no membrane.
Prokaryotes have no organelles at all.	They have ribosomes, but no membrane-bound organelles.
The nucleus has a single membrane.	The nuclear envelope is a double membrane with pores.
Nuclear pores let DNA out.	DNA stays in; mRNA leaves.

Check that you can...

1. **Define** an organelle and **classify** structures by their membranes
2. **Outline** the jobs of the main organelles
3. **Explain** the advantages of separating the nucleus from the cytoplasm
4. **Explain** the advantages of compartments in the cytoplasm
5. **Describe** how organelles work together to secrete a protein

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Define an organelle and give one example with two membranes.
2. Explain one advantage of separating the nucleus from the cytoplasm.
3. Explain why lysosome enzymes are kept in a compartment.

Image credits

Mammalian pancreatic tissue, nuclear pores and rough endoplasmic reticulum.jpg – Louisa Howard – Public domain

Animal cell structure en.svg – LadyofHats (Mariana Ruiz) – Public domain

Average prokaryote cell- en.svg – Mariana Ruiz Villarreal, LadyofHats – Public domain

Mammalian pancreatic acinar cell, nuclear pores -TEM.jpg – Louisa Howard – Public domain

Mitochondrion structure.svg – Kelvinsong; modified by Sowlos – CC BY-SA 3.0

Chloroplast II.svg – Kelvinsong – CC BY 3.0

Golgi apparatus (standalone version)-en.svg – Kelvinsong – CC BY 3.0

Paramecium caudatum.jpg – Alfred Kahl – Public domain

Endocytosis types.svg – Mariana Ruiz Villarreal LadyofHats – Public domain

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